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Roll No

EE-7004(1)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Advance Power Electronics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What is the purpose of resonant converters in the design of SMPS? Explain any one technique.
b) What is the difference between linear and switched mode converters give one example of each converter?
2. Explain the following terms related to power electronics supply
 - i) Bulk capacitor
 - ii) Hold up time
 - iii) Inrush current
 - iv) Current limiting resistor
 - v) ESR
 - vi) Line regulation
 - vii) Load regulation
3. A simple step down DC-DC converter has a resistive load of $10\ \Omega$ and the input voltage of 220 volt when the converter switch remains on, its voltage drop is 2V and the chopping frequency is 1 kHz. If duty cycle is 50%, determine-
 - i) Average output voltage
 - ii) RMS output voltage
 - iii) Converter efficiency
 - iv) Effective input resistance of the converter
 - v) RMS value of the fundamental component of output voltage.

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4.
 - a) Derive expressions for minimum and maximum values of load current in a step down chopper with R-L-E load.
 - b) An RLE load is operating in AC chopper circuit from a 500 volts dc source. For the load, $L = 0.06H$, $R = 0$ and constant E . For a duty cycle of 0.2 find the chopping frequency to limit the amplitude of load current excursion to 10A.
5.
 - a) Explain three phase 180° mode VSI with appropriate wave form and circuit diagrams.
 - b) A three phase bridge inverter delivers power to a resistive load from a 450v dc source. For a star-connected load of 10 ohm per phase, determine for both
 - i) 180° mode and
 - ii) 120° mode.
 - 1) .rms value of load current
 - 2) .rms value of thyristor current
 - 3) .load power
6.
 - a) With the help of neat circuit diagram and waveforms for all gating signals of the switches, all phase voltages and all line to line voltages explain the operation of 120° deg. conduction for three phase inverter connected to balance Y- connected. Give the expression for line to line voltage.
 - b) Using proper circuit diagram and waveforms. Explain the operation of three phase current source inverter.
7.
 - a) Classify and explain various AC choppers using circuit topology.
 - b) Explain the symmetrical and asymmetrical patterns for harmonic control in AC choppers.
8.
 - a) What is soft switching in DE DC converter. Explain with an example for buck converter.
 - b) Explain with proper diagram and waveform ZVS operation for boost converter.

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